CARMEN-Q

Causal Audit–Return Duality for Streaming Quantum Memory

An exact classical-memory frontier and a coherent one-qubit separation

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Abstract

Can a device remember a property of a quantum stream and still return the stream coherently? This work turns that question into a two-branch benchmark. A trusted verifier sends one half of each of n independent EPR pairs through a device and sequesters every output before releasing the next input. Only after the terminal memory is committed does a hidden choice request either the parity of later reference measurements (AUDIT) or recovery of every EPR pair together with visible memory reset (RETURN). For strategies comprising arbitrary adaptive local instruments with unlimited classical memory but no coherent state persisting between slots, the exact support function is

$$f(t) = \frac{1+\sqrt{1-t^2}}{2},$$
$$C^{\text{stream}}_{n,\lambda} = \max_{0\le t\le 1}\left\{\frac{\lambda}{2}(1+t^n) + (1-\lambda)f(t)^n\right\}.$$

With equal branch weights, this classical-memory ceiling is 3/4 for every $n \ge 2$, while one persistent coherent qubit reaches $(P_{\mathrm{A}}, F_{\mathrm{R}}) = (1, 1)$. Removing the streaming restriction raises the classical ceiling, which isolates the operational cost of causality. The analysis also identifies a first-order transition in the optimal classical strategy for $n \ge 3$, supplies a fixed-sample certificate, and specifies a hardware-oriented protocol. The result is a trusted-interface temporal-memory witness, not a device-independent claim or a proof of global erasure.

Keywords: quantum memory; multitime quantum processes; information–disturbance; entanglement fidelity; reversible computation; causal benchmarks.

1 Introduction

When does a temporal memory have to be quantum? A final interference fringe does not answer the question: a known circuit can sometimes be replaced by an endpoint-equivalent channel that never stores the advertised intermediate record. A useful test must instead force one committed process to support two incompatible later uses.

The protocol in Fig. 1 implements that idea. A verifier sends fresh entangled carriers through a device one at a time, removing each returned carrier before the next one arrives. After the device commits its terminal memory, a hidden coin selects between two branches. AUDIT asks for the parity of later measurements on the verifier's reference systems, whereas RETURN asks the device to help restore every entangled pair and reset the visible memory port. Learning the stream helps AUDIT but tends to damage RETURN; doing nothing protects RETURN but leaves parity at chance. A coherent qubit can do both because it accumulates parity reversibly and is later either read or uncomputed.

At a glance. At balanced branch weight, every allowed adaptive classical-memory strategy scores at most 0.75 for $n \ge 2$. One coherent memory qubit scores 1. The gap is the experimentally testable result.

The constituent ideas are established: information acquisition competes with disturbance [1]; path-versus-phase games use late choices [2]; parity discrimination appears in quantum cryptography [3]; and process tensors classify temporal resources [4, 5, 6]. The contribution is a precise streaming interface together with an exact optimisation over the full adaptive classical-memory null.

The proof has a simple backbone. First, a recovery lemma turns any local instrument, including non-QND dynamics, into a bound expressed only through its classical likelihoods. Causal factorisation then makes both parity bias and recoverable entanglement multiply along a transcript. Finally, convexity shows that equal local measurement strengths are optimal. This yields an exact one-variable frontier rather than a numerical bound or one based on a symmetric ansatz.

The same task exposes three resource classes. Streaming classical memory is weakest. Collective processing of all carriers before retaining a classical record is stronger. A persistent coherent qubit reaches the algebraic opti-

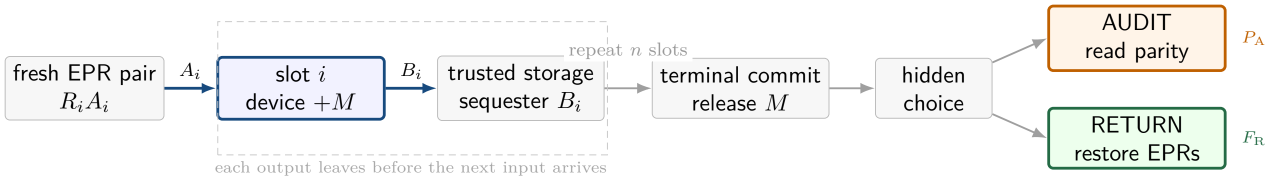

Figure 1: The benchmark as a causal story. The verifier releases one EPR half at a time and sequesters each output before the next slot. Only after the terminal memory is committed does the hidden challenge choose between reading a temporal predicate (AUDIT) and reversing the disturbance (RETURN).

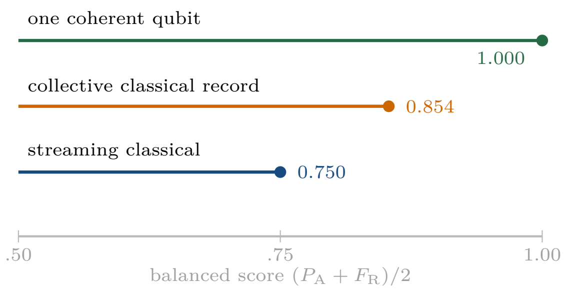

Figure 2: Balanced score for the three comparison classes when $n \geq 2$. Streaming classical memory is capped at $3/4$; collective classical recording reaches $(1 + 1/\sqrt{2})/2$; a coherent parity accumulator reaches unity.

mum. Figure 2 previews this separation; the exact statements follow in Section 4.

The benchmark is deliberately model-relative. It trusts the source, timing, sequestration, branch measurements, and the declared boundary around cross-slot memory. A positive experiment rejects that classical-memory model; it does not identify a unique microscopic implementation, prove deletion from an inaccessible environment, or select an interpretation of quantum mechanics.

1.1 Contributions and reading guide

The main results are the exact streaming frontier, a strict collective-classical comparator, a perfect one-qubit strategy, and the first-order transition in the optimal classical strategy. Section 3 defines the game, Section 4 states the mathematics, and Sections 5 and 7 develop the experimental test. Proofs and reproducibility details are deferred to the appendices so the main argument can be read independently of them.

2 Scope and related work

The closest literature supplies the ingredients of the benchmark, but not the complete streaming task. The distinctions below are stated explicitly because the comparison class is part of the result.

2.1 The comparison class is the claim

Quantum memory has several inequivalent meanings in a multitime process. Process tensors and quantum combs provide the general slot structure [4, 5]. Memory cost can be a global property of a strategy even when classical assistance is free [7]. Recent classifications distinguish processes that transmit coherent quantum systems, classical information through conditional instruments, or no information at all [6]. This work uses a direct operational restriction: after refining every outcome and shared-randomness seed, each leaf CP map factorises over slots. Disposable quantum ancillas are allowed inside a slot; no coherent state or entangled common cause persists between slots.

This condition is not equivalent to entanglement breaking of any reduced channel. It is weaker than memorylessness because it retains unlimited adaptive classical state. Arbitrary finite alphabets, non-QND maps, and transcript-conditioned joint recovery are also included. A positive experiment is therefore a resource witness only relative to trusted timing, source, sequestration, and terminal-port assumptions.

2.2 Information–disturbance and operational duality

One-shot estimation–operation fidelity tradeoffs establish that extracting classical information disturbs quantum states [1]. Operational wave–particle duality similarly frames a late game between path and phase information [2]. Quantum seals study how readability limits later verification of an undisturbed state [8]. The one-slot curve used here is not new; it is the binary information–return relation. The contribution is the causal tensorisation of that relation for fresh sequestered inputs, coupled to a parity audit of the retained temporal record and a global return score for all carriers under the full adaptive classical-memory null.

Parity-based discrimination is also established territory. Collective measurements can reveal global parity information unavailable to separate measurements in quantum cryptographic ensembles [3]. Here parity is chosen because a coherent accumulator needs only one qubit while any classical streaming strategy multiplies local posterior biases. The task is not cryptographic and does not claim secrecy.

2.3 Temporal-memory witnesses

Memory witnesses based on multitime process entanglement, effective environment dimension, and identity-channel behaviour already exist [9, 10, 11]. Sequential random-access codes can separate classical and quantum temporal memory under semi-device-independent assumptions [12]. Bounded coherent and classical memory in adaptive discrimination can be formulated through constrained separability [13]. The score used here differs because its second branch asks for joint entanglement return of the same committed carriers, rather than a later decoding success. This is an adjacent benchmark, not a replacement for those frameworks.

2.4 Reversible computation and interpretation

The honest strategy uses CNOT accumulation and uncomputation, both standard tools of reversible computing [14]. Its value comes from the late branch and access constraints, not from describing the circuit as multiple histories. The observed probabilities are fully predicted by standard quantum mechanics. No interpretation-specific claim follows.

2.5 Priority discipline

An adversarial search through 12 August 2026 found no primary source stating the full theorem with all of the following: fresh entangled inputs, immediate sequestration, adaptive non-QND instruments with only classical inter-slot memory, late transcript-only parity audit, and joint all-carrier return. A literature search cannot prove absence; the priority claim should be withdrawn if an earlier theorem specialises directly to this interface and scoring rule.

3 Operational model

This section turns the story in Fig. 1 into a score. Fix $n \geq 1$. At slot i, a trusted verifier prepares

$$|\Phi\rangle_{R_iA_i} = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

sends A_i to the device, receives B_i, and sequesters B_i before releasing A_{i+1}. After the last output, the device commits a terminal memory port M to the verifier. A hidden coin then selects one branch.

AUDIT. The verifier measures every R_i in the Z basis, obtaining independent uniform bits X_i, and uses M to decode a guess for

$$Y = X_1 \oplus \cdots \oplus X_n.$$

The sequestered B_i are unavailable. Let P_A be the success probability.

RETURN. The verifier applies a fixed joint decoder to $MB_1 \cdots B_n$ and accepts with

$$\Pi_\mathrm{R} = |0\rangle\langle 0|_M \otimes \bigotimes_{i=1}^{n} |\Phi\rangle\langle\Phi|_{R_iB_i}.$$

Let F_R be the acceptance probability. The visible reset $M = 0$ does not imply deletion of any inaccessible purification of a classical record.

For $0 \leq \lambda \leq 1$, define the support score

$$S_{n,\lambda} = \lambda P_\mathrm{A} + (1-\lambda) F_\mathrm{R}.$$

The balanced benchmark uses $\lambda = 1/2$.

3.1 Adaptive classical-memory streaming null

The comparison class is intentionally powerful: it may learn, randomise, and adapt without bound. The single excluded resource is a coherent state that survives from one slot to the next.

The null may use arbitrary finite classical memory and shared randomness. In slot i, conditional on the earlier classical history $h = k_{<i}$, it applies an arbitrary instrument

$$\{\mathcal{N}_{k_i}^{(i,h)} : A_i \to B_i\}_{k_i}.$$

It may create and discard arbitrary within-slot quantum ancillas. It may not retain a coherent system across the slot boundary, begin with entanglement shared among ancillas assigned to different slots, revisit a committed B_i, or receive a B_i in AUDIT. In RETURN, the full classical transcript may control an arbitrary joint recovery on all sequestered outputs.

After refining all classical random seeds and outcomes, every leaf transcript $k = (k_1, \ldots, k_n)$ induces a trace-nonincreasing CP map that factorises over slots. For computational-basis inputs $x = (x_1, \ldots, x_n)$, its likelihood is

$$p(k|x) = \prod_{i=1}^{n} p_i(k_i|x_i, k_{<i}). \tag{1}$$

This factorisation is the mathematical null. Timing and isolation evidence must justify it; the score alone cannot enforce a laboratory boundary.

3.2 Local audit strength and return curve

Every local outcome can now be summarised by one number: how strongly it favours input 0 over input 1. This makes the information–disturbance tradeoff visible without committing to a particular measurement implementation.

At a realised prefix h, define

$$r_i(k_i|h) = \frac{p_i(k_i|0,h) + p_i(k_i|1,h)}{2}$$

and, when $r_i > 0$, define $\delta_i \in [0,1]$ and $s_i \in \{-1, 1\}$ by

$$s_i\delta_i = \frac{p_i(k_i|0,h) - p_i(k_i|1,h)}{p_i(k_i|0,h) + p_i(k_i|1,h)}, \qquad 0 \leq \delta_i \leq 1. \tag{2}$$

Table 1: Nearest occupied territory and the remaining distinction.

Area	Established result	Distinction here
Information–disturbance	Tight one-shot fidelity tradeoffs	Exact causal product law for a parity audit and global return
Ways/Phases games	Late operational duality choices	Fresh time slots, sequestered outputs, and retained temporal memory
Parity discrimination	Collective advantage for global parity	Shared-prefix global-return score and a classical-memory comb null
Quantum seals	Readability versus later verification	Multitime stream and all-carrier EPR return
Temporal memory witnesses	Classical/quantum hierarchy and sequential scores	Exact same-task support and attainment with one coherent qubit
Reversible computing	Compute–uncompute circuits	Hidden late challenge and a falsifiable resource separation

Either sign may be chosen when $\delta_i = 0$. The one-slot return curve is

$$f(t) = \frac{1+\sqrt{1-t^2}}{2}. \tag{3}$$

Here $t=0$ extracts no computational-basis information and preserves an EPR pair perfectly; $t=1$ permits a perfect bit record and leaves entanglement fidelity $1/2$.

3.3 A saturating null family

The following weak measurement saturates the one-slot return curve. It interpolates continuously between doing nothing ($t_i = 0$) and making a projective record ($t_i = 1$).

For any $t_i \in [0,1]$, consider the binary diagonal instrument

$$K_0^{(i)} = \mathrm{diag}\left(\sqrt{\frac{1+t_i}{2}}, \sqrt{\frac{1-t_i}{2}}\right),$$
$$K_1^{(i)} = \mathrm{diag}\left(\sqrt{\frac{1-t_i}{2}}, \sqrt{\frac{1+t_i}{2}}\right).$$

It produces local audit bias t_i and return fidelity $f(t_i)$. The transcript parity is the optimal AUDIT guess and identity recovery saturates the RETURN expression derived below.

4 Exact frontiers and resource separation

The full frontier is stated first, followed by its consequences. Readers interested mainly in the experimental message can focus on the highlighted balanced case and the resource comparison; all proof details are in Appendix A.

Theorem 4.1 (Streaming parity audit–return frontier)**.** *For the adaptive classical-memory null in Section 3, including arbitrary non-QND local instruments and arbitrary transcript-conditioned joint recovery,*

$$C^{\mathrm{stream}}_{n,\lambda} = \max_{0\le t_1,\dots,t_n\le 1}\left[\frac{\lambda}{2}\left(1+\prod_{i=1}^{n} t_i\right) + (1-\lambda)\prod_{i=1}^{n} f(t_i)\right] \tag{4}$$

$$= \max_{0\le t\le 1}\left[\frac{\lambda}{2}(1+t^n) + (1-\lambda)f(t)^n\right]. \tag{5}$$

The bound is attainable by independent weak diagonal instruments.

The proof has three moves. First, a Choi-positivity lemma bounds recoverable EPR fidelity using only basis-state likelihoods. Second, causality makes both the posterior parity bias and the return fidelity factorise across slots. Third, convexity shows that an optimal strategy uses the same strength in every slot. This is why the many-instrument optimisation collapses to the single variable t in (5).

Corollary 4.2 (Balanced null)**.** *For $n \ge 2$ and $\lambda = 1/2$,*

$$C^{\mathrm{stream}}_{n,\lambda} = \frac{3}{4}.$$

The least-disturbing optimum is $t=0$: the null learns no parity beyond chance and returns every EPR pair.

At a glance. At $\lambda = 1/2$, a classical-memory device cannot improve the score by measuring weakly: for every $n \ge 2$, its best choice is to keep no record, guess parity, and return the EPR pairs. The ceiling is therefore $\frac{1}{2}(\frac{1}{2}+1) = \frac{3}{4}$.

4.1 Closed forms and the strategy transition

For one slot,

$$C^{\mathrm{stream}}_{1,\lambda} = \frac{1}{2} + \frac{1}{2}\sqrt{\lambda^2 + (1-\lambda)^2}.$$

For two slots,

$$C^{\mathrm{stream}}_{2,\lambda} = \begin{cases} 1-\lambda/2, & 0\le\lambda\le 1/2, \\ \lambda/2 + \lambda^2/(3\lambda-1), & 1/2<\lambda\le 1. \end{cases} \tag{6}$$

The optimal strength grows continuously from zero at $\lambda = 1/2$.

The weighted game reveals additional structure. For $n \geq 3$, the optimal classical strategy changes discontinuously as AUDIT becomes more important. Let z_n be the unique solution

$$(1 - z_n)(1 + z_n)^{n-1} = 1, \qquad z_n \in \left(\frac{n-2}{n}, 1\right). \quad (7)$$

Then

$$\lambda_c(n) = \frac{1}{1 + 2^{n-2} z_n^{(n-2)/2}(1 - z_n)}. \quad (8)$$

Below λ_c, the optimum is $t = 0$. At λ_c, $t = 0$ ties a finite-strength branch, so the optimal strength jumps discontinuously. For $n = 3, 4, 5$, respectively, (λ_c, t_c) is approximately $(0.6248, 0.9717)$, $(0.6495, 0.9962)$, and $(0.6589, 0.9993)$. Moreover, $\lambda_c(n) \to 2/3$ from below. The optimal null thus changes from recording nothing to recording each slot almost projectively rather than moving through a broad weak-measurement phase.

4.2 Collective classical-record comparator

Now remove streaming sequestration and allow one joint instrument on all carriers, while requiring that the device retain only classical side information after commitment. The quantum output carriers remain in trusted storage and are available to the RETURN decoder.

Proposition 4.3 (Collective classical record)**.** *For every n,*

$$C_\lambda^{\text{coll}} = \frac{1}{2} + \frac{1}{2}\sqrt{\lambda^2 + (1 - \lambda)^2}. \quad (9)$$

For $n \geq 2$ and $0 < \lambda < 1$, this is strictly larger than $C_{n,\lambda}^{\text{stream}}$.

A weak measurement of the two parity subspaces attains (9). The device still retains only a classical outcome; its advantage comes from seeing all carriers together. The comparator therefore isolates the advantage lost when the apparatus must operate as a stream.

4.3 One coherent qubit closes the gap

Initialise a memory qubit $M = |0\rangle$. In slot i, apply $\text{CNOT}_{A_i \to M}$ and return A_i. After the final slot, commit M. In AUDIT, measuring M in Z gives the parity of the later Z measurements on the references. In RETURN, applying $\text{CNOT}_{B_i \to M}$ for every sequestered carrier uncomputes M to $|0\rangle$ and exactly restores all EPR pairs. Thus $(P_{\text{A}}, F_{\text{R}}) = (1, 1)$. The operational resource is a coherent temporal accumulator, not an observer or an interpretation of quantum mechanics.

5 Statistical certificate and experimental protocol

The theorem gives a probability bound; an experiment needs a decision rule. The rule below separates three ingredients that are often conflated: sampling uncertainty, measured bias in each branch, and uncertainty in whether the laboratory really implements the theoretical null.

5.1 Fixed-sample certificate

Let K_{A} and K_{R} denote the numbers of successes, and let N_{A} and N_{R} denote the prespecified trial counts in the two branches. Define

$$\widehat{S} = \lambda \frac{K_{\text{A}}}{N_{\text{A}}} + (1 - \lambda)\frac{K_{\text{R}}}{N_{\text{R}}}.$$

For independent Bernoulli trials, the one-sided weighted Hoeffding radius is

$$r_\alpha = \sqrt{\frac{1}{2}\left(\frac{\lambda^2}{N_{\text{A}}} + \frac{(1-\lambda)^2}{N_{\text{R}}}\right)\log\frac{1}{\alpha}}. \quad (10)$$

Before observing outcomes, let δ_{A} and δ_{R} upper-bound possible upward bias in the two estimators, and let δ_{null} enlarge the theoretical null for source, isolation, and interface uncertainty. Incompatibility with the declared classical-memory null is certified only if

$$\widehat{S} - \lambda\delta_{\text{A}} - (1 - \lambda)\delta_{\text{R}} - r_\alpha > C_{n,\lambda}^{\text{stream}} + \delta_{\text{null}}. \quad (11)$$

Under independent fixed-sample trials, the false-positive probability is at most α. Choosing n or λ after seeing data, optional stopping, or selecting the best of several lengths requires a different analysis.

For an alternative score S_{alt}, define

$$g = S_{\text{alt}} - \lambda\delta_{\text{A}} - (1 - \lambda)\delta_{\text{R}} - C_{n,\lambda}^{\text{stream}} - \delta_{\text{null}}.$$

If $g \leq 0$, no amount of data certifies the claim under those allowances. If $g > 0$, fixed sample sizes satisfying $r_\alpha + r_\beta < g$ guarantee power at least $1 - \beta$ by a second Hoeffding bound. For a fixed total trial budget, the asymptotically optimal allocation is $N_{\text{A}} : N_{\text{R}} = \lambda : (1 - \lambda)$.

5.2 Commitment and branch timing

A valid trial follows the left-to-right order of Fig. 1. The verifier first prepares and independently characterises the EPR sources. It releases A_i only after B_{i-1} has entered trusted sequestration and, after receiving B_n, requires delivery of the terminal memory port M. Only then is the hidden AUDIT/RETURN choice generated. The preregistered branch decoder and binary acceptance measurement are applied, and every loss or exclusion is recorded under a predeclared rule. Branch assignments may be randomised across trials, but the branch sample sizes and stopping rule must be prespecified or handled by a sequentially valid analysis.

5.3 Mandatory shortcut controls

These controls are part of the scientific claim, not optional diagnostics. Deliberately dephasing the persistent carrier should remove the coherent advantage, whereas making a strong classical record should improve AUDIT while lowering global RETURN. Removing sequestration should recover the collective curve in (9); allowing the device to revisit an earlier B_i should likewise expose why the causal boundary matters. Decoder controls replace the coherent inverse by identity or by a mismatched inverse. Independent source and measurement calibrations must bound false EPR acceptance and parity bias, and a blind-choice control must test whether branch information leaks before commitment. Finally, a side-channel audit must bound coherent transfer through buses, pulse tails, clocks, resonators, unused levels, and shared ancillas.

The null does not forbid inaccessible erasure environments. Consequently, visible reset of M is a reproducibility and wiring check, not a certificate that every physical copy of a classical transcript has vanished.

5.4 Reproducible analysis interface

The public command-line tool evaluates the exact bound, plans trial counts, and applies the fixed-sample rule through three commands with strict JSON output:

```
carmenq bound --steps 8
carmenq plan --steps 8 --forecast-model
carmenq analyse --steps 8 \
  --audit-successes 9700 --audit-trials 10000 \
  --return-successes 9500 --return-trials 10000
```

Systematic allowances, α, β, and null slack are command-line inputs and appear in the output. An example preregistration records the terminal port, source calibration, decoder, exclusions, controls, and stopping rule.

6 Numerical validation and forecast

The theorem is analytic. Computation plays two supporting roles: it searches for counterexamples to the derivation and translates an explicit noise assumption into an experimental scale. It does not substitute for the proof.

6.1 Adversarial theorem checks

Independent scripts checked the one- and two-slot closed forms and the transition equations through $n = 12$. They also tested random product likelihoods against the flagged entanglement-recovery inequality, random collective likelihood tables against (9), and the exact coherent-accumulator circuit. The most adversarial searches allowed asymmetric local strengths and fully adaptive binary-outcome likelihood trees for $n = 2, \ldots, 5$. The maximum numerical excess over the streaming theorem was 6.52×10^{-14} for adaptive trees and 3.53×10^{-14} for asymmetric products, consistent with numerical precision. The smallest tested collective-classical gap was positive. These checks do not prove the theorem or priority; they are regression tests against algebraic and implementation errors.

The release suite contains 26 automated tests covering the exact supports, transition behaviour, systematic allowances, sample planning, deterministic simulation, and reproducible output.

6.2 Transparent forecast model

For planning only, the forecast uses the replaceable phenomenological law

$$P_{\mathrm{A}}(n) = \frac{1 + c_0 c^n}{2}, \qquad F_{\mathrm{R}}(n) = f_0 f^n. \tag{12}$$

The versioned defaults are $c_0 = f_0 = 0.995$, $c = 0.998$, and $f = 0.997$. They are illustrative logical-operation factors, not hardware calibration.

With $\lambda = 1/2$, $\alpha = 0.01$, target power 0.9, 0.005 systematic allowance in each branch, and 0.005 null slack, the model remains certifiable through $n = 143$ and fails at $n = 144$. Its unadjusted score crosses the 3/4 null at $n = 151$. As the robust margin closes, the conservative shot count grows rapidly.

At $n = 8$, a declared alternative $(P_{\mathrm{A}}, F_{\mathrm{R}}) = (0.97, 0.95)$ under the same allowances needs 84 trials per branch for the Hoeffding guarantee ($\alpha = 0.01$, power at least 0.9). If 9700 of 10000 AUDIT trials and 9500 of 10000 RETURN trials succeed, the adjusted one-sided lower score is approximately 0.9443, exceeding the enlarged 0.755 null by approximately 0.1893.

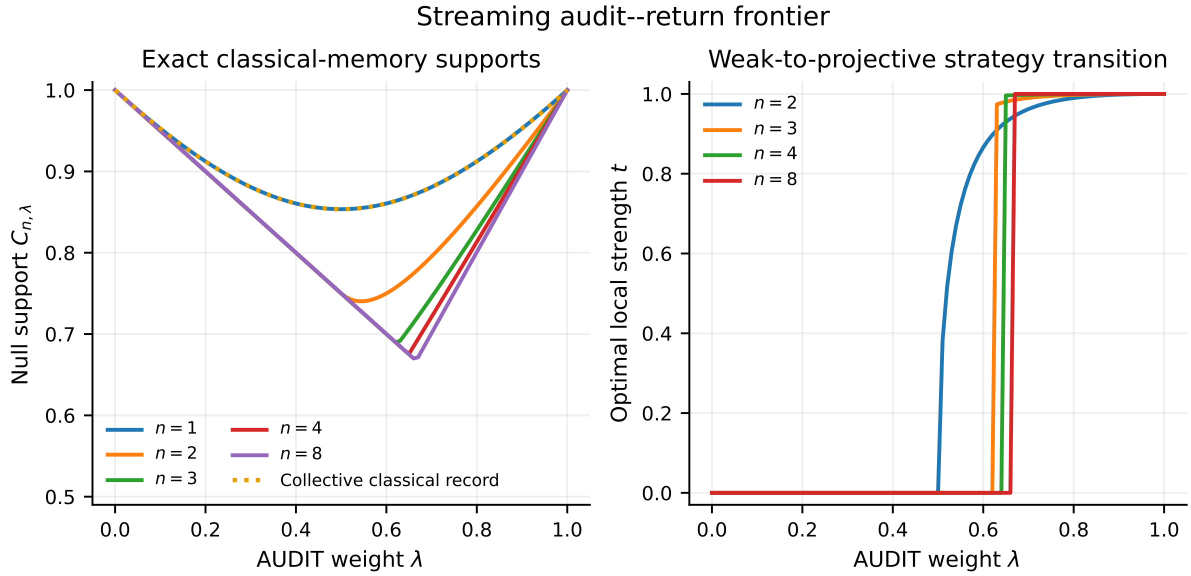

Figure 3: Exact streaming supports and exposed local strengths. At $n \geq 3$, the optimal classical strategy jumps from no record to an almost projective record as the audit weight crosses $\lambda_c(n)$. The dotted curve is the collective classical-record comparator, not an unrestricted coherent strategy.

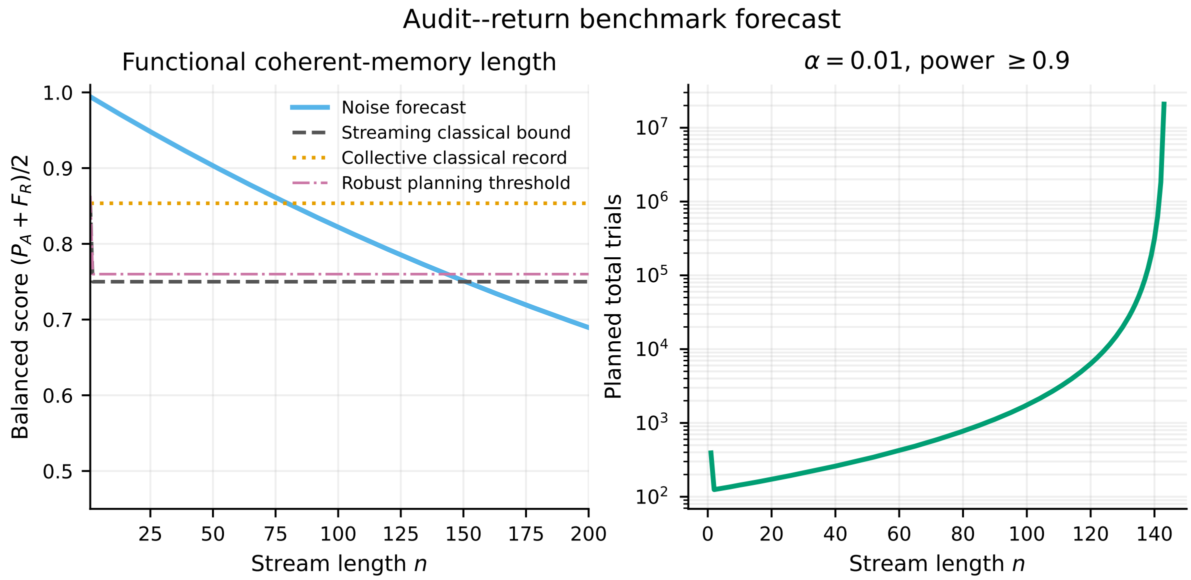

Figure 4: Illustrative forecast from (12). Left: balanced score against the streaming, collective-classical, and robust planning thresholds. The one-slot streaming point is higher than the $n \geq 2$ plateau. Right: conservative total trial count from (10); infeasible lengths are omitted. These curves are not hardware predictions.

6.3 Functional coherent-memory length

The *certifiable functional coherent-memory length* is defined as the largest preregistered n for which (11) is satisfied under a declared familywise testing rule. Unlike a bare T_2, it asks a memory to retain a temporal predicate and later participate in coherent global return. It remains benchmark-specific: changing the predicate, branch scores, or sequestration interface changes the resource being measured.

7 Experimental feasibility

The logical circuit is small. The demanding part is proving that information could not travel around the declared memory boundary.

7.1 Minimal logical circuit

The honest coherent device needs one persistent qubit and n streamed carrier interactions. The source first creates n EPR pairs. In each slot the device applies $\mathrm{CNOT}_{A_i \to M}$, returns the carrier, and a trusted switch routes the resulting B_i to inaccessible storage. Once the terminal memory is committed, AUDIT measures M and the R_i systems in Z. RETURN instead applies the n inverse CNOTs from the stored B_i systems to M, Bell-tests every R_iB_i pair, and finally tests $M = 0$. Thus the untrusted device uses one two-qubit gate per slot, while the recovery station adds n inverse gates and n Bell analysers. The global RETURN event is deliberately stringent: every pair and the terminal reset must pass in the same trial.

7.2 The hard part is the interface

An implementation needs a switch or transport geometry demonstrating that B_i left the device before A_{i+1} arrived, together with sufficiently quiet storage for the later joint decoder. The terminal port must transfer M without revealing the branch choice, while calibration must suppress coherent paths that cross slot boundaries outside the memory counted by the model. The branch generator and its control electronics must likewise be timed so that they cannot signal the future choice to the device. A hidden bus mode or coherent pulse tail is not merely noise under the theorem; it is an unaccounted quantum-memory resource. It may still produce a positive witness of *some* coherent memory, but it prevents attribution to the declared qubit.

7.3 Candidate platforms

Superconducting circuits can combine fast controlled gates, a dedicated memory or bus mode, and microwave routing, but resonator and control-line remnants require an explicit side-channel budget. Trapped ions offer long-lived internal memories and high-quality entangling gates, while physical sequestration is naturally expressed through shuttling, zones, or trusted spectator isolation. Neutral-atom arrays offer reconfigurable geometry and parallel entanglement tests, provided that loss treatment is preregistered. Photonic time bins coupled to a matter memory naturally realise a stream, but source loss, storage loss, and postselection are central risks.

Headline gate fidelity is therefore not enough. The decisive platform property is whether the causal interface can be made credible and independently audited.

7.4 Staged demonstration

A staged demonstration should begin at $n = 2$, where the coherent point can be compared directly with a dephased-memory control. The next stage implements the weak diagonal instruments and traces the classical frontier as λ varies. Enforced streaming can then be compared with collective access on the same hardware, followed by preregistered tests at multiple stream lengths using simultaneous confidence intervals or multiplicity-corrected tests. An independent interface audit should characterise leakage channels without using the primary score data.

7.5 Kill criteria

The flagship claim must be reduced or abandoned if the classical-path factorisation fails to match the experimental null, or if an adaptive classical strategy exceeds (5). The same applies if the branch choice or sequestered carriers become accessible before commitment, if postselection or loss rules condition the score on hidden outcomes, or if source correlations invalidate the assumed independence of the X_i. Replacing the global RETURN event with average pair fidelity also changes the theorem. Finally, a direct prior theorem would remove the novelty claim, while systematic allowances large enough to erase the gap would remove the experimental conclusion. Any of these outcomes could still leave useful calibration or pedagogy, but not the exact resource witness stated here.

8 Discussion

The benchmark turns an imprecise question about reversible records into one number with an exact null. AUDIT alone rewards destructive measurement; RETURN alone rewards doing nothing. A hidden late choice forces one committed process to remain useful for both, while sequestration prevents a classical device from replacing the stream with one global parity measurement.

The first-order transition comes from parity. Weak local evidence combines multiplicatively, so a modest per-slot record becomes exponentially poor at predicting a long parity. Return fidelity also multiplies. For $n \geq 3$, the optimum therefore skips a range of intermediate strategies and jumps from no measurement to almost projective measurement. This is a transition in an optimisation problem, not in matter.

The collective comparator shows why timing matters: the same classical record becomes more powerful when all carriers are jointly accessible. The coherent strategy also shows that memory dimension need not grow with the stream; a single protected qubit stores parity ideally, although global return becomes more fragile as n grows.

8.1 Limitations

The central limitations are about trust and attribution.

The benchmark is not device independent. It trusts the EPR source, the branch measurements, the sequestration station, the timing boundary, and the absence or calibrated allowance of unaccounted coherent side channels. It certifies incompatibility with a declared classical-memory comb; it does not reconstruct the device's internal dynamics or prove that the named qubit was the only memory.

Visible terminal reset is not global erasure. A classical controller may reset its public port while dumping its transcript into an inaccessible environment. The present null allows this and the theorem remains valid. Demanding decoupling from every purification would define a different, stronger task.

The finite-statistics result assumes independent Bernoulli trials and fixed sample sizes. Drift, correlated errors, adaptive stopping, multiple lengths, and estimated systematic bounds need dedicated inference. The phenomenological forecast is a planning example, not a hardware claim.

No interpretation of quantum mechanics is selected. A successful result says that a coherent temporal resource outperforms an explicit classical-memory model on an ordinary quantum-information task.

8.2 Extensions

The most direct theoretical extensions are exact frontiers for q-ary streams and predicates beyond parity, followed by tight bounds for finite coherent-memory dimensions and robust nulls with bounded cross-slot leakage. On the statistical side, optional stopping and adaptive length selection call for sequentially valid inference. A stronger certification programme would pursue semi-device-independent or device-independent variants with operationally certified sequestration. Experimentally, the clearest next comparison is streamed versus collective access using the same physical components.

8.3 Conclusion

For a streamed parity task, adaptive classical memory obeys an exact audit–return frontier, collective classical processing does strictly better, and one coherent temporal qubit reaches the algebraic optimum. The remaining challenge is experimental: make the causal boundary as credible as the mathematical one.

A Proofs

A.1 Flagged entanglement-recovery lemma

Lemma A.1. *Let $\mathcal{N}_k : A \to B$ be a trace-nonincreasing CP map, $\dim A = D$, and*

$$p(k|x) = \operatorname{Tr} \mathcal{N}_k(|x\rangle\langle x|)$$

in a fixed orthonormal basis. For any recovery channel $\mathcal{R}_k : B \to A$, its unnormalised contribution to the maximally entangled-state fidelity satisfies

$$F_k \le \frac{1}{D^2}\left(\sum_x \sqrt{p(k|x)}\right)^2. \tag{13}$$

The bound is attainable for every valid likelihood table by a positive diagonal instrument.

Proof. Put $\Lambda_k = \mathcal{R}_k \circ \mathcal{N}_k$. Expanding the entanglement fidelity with $|\Phi_D\rangle = D^{-1/2}\sum_x |x\rangle_R|x\rangle_A$ gives

$$D^2 F_k = \sum_{x,y} \langle x|\Lambda_k(|x\rangle\langle y|)|y\rangle.$$

Positivity of the Choi matrix implies

$$\begin{aligned}|\langle x|\Lambda_k(|x\rangle\langle y|)|y\rangle| &\le \Big[\langle x|\Lambda_k(|x\rangle\langle x|)|x\rangle\langle y|\Lambda_k(|y\rangle\langle y|)|y\rangle\Big]^{1/2}\\ &\le \sqrt{p(k|x)p(k|y)}.\end{aligned}$$

Summing proves (13). For tightness, choose $K_k = \sum_x \sqrt{p(k|x)}|x\rangle\langle x|$ and identity recovery. Completeness of the likelihood table ensures $\sum_k K_k^\dagger K_k = \mathbb{I}$. □

A.2 Causal tensorisation

For a refined transcript k and the definitions in (2), every fresh X_i is uniform and independent of the previous transcript. Bayes' rule and (1) therefore make the posterior distribution of $X_1, \dots, X_n$ a product along the realised leaf. The conditional parity bias is

$$\left|\mathbb{E}[(-1)^{X_1+\cdots+X_n}|k]\right| = \prod_i \delta_i.$$

Averaging the optimal parity guess gives

$$P_{\mathrm{A}} = \frac{1}{2}\left(1 + \mathbb{E}_k \prod_i \delta_i\right). \tag{14}$$

Condition on the refined classical transcript. Dropping the terminal reset projector can only increase RETURN acceptance, leaving a transcript-controlled recovery channel on the carrier block. Applying (13) with $D = 2^n$, using (1), and summing independently over the input strings at each slot gives

$$\begin{aligned}F_{\mathrm{R}} &\le 4^{-n}\sum_k\left(\sum_x \sqrt{p(k|x)}\right)^2 \qquad (15)\\ &= \sum_k \prod_i \frac{\left(\sqrt{p_i(k_i|0,h)} + \sqrt{p_i(k_i|1,h)}\right)^2}{4} \qquad (16)\\ &= \mathbb{E}_k \prod_i \frac{1+\sqrt{1-\delta_i^2}}{2} = \mathbb{E}_k \prod_i f(\delta_i). \qquad (17)\end{aligned}$$

Thus

$$S_{n,\lambda} \le \frac{\lambda}{2} + \mathbb{E}_k\left[\frac{\lambda}{2}\prod_i \delta_i + (1-\lambda)\prod_i f(\delta_i)\right].$$

An expectation cannot exceed the largest realised bracket, proving the upper bound in (4). The diagonal binary family in Section 3 attains every deterministic strength profile, proving tightness and showing that neither adaptivity nor non-QND dynamics can improve the support.

A.3 Equal strengths

Set

$$t_i = \frac{2y_i}{1+y_i^2}, \qquad f(t_i) = \frac{1}{1+y_i^2}, \qquad 0 \le y_i \le 1.$$

Apart from the additive $\lambda/2$, the objective is

$$G(y_1, \dots, y_n) = \frac{(1-\lambda) + \lambda 2^{n-1}\prod_i y_i}{\prod_i (1+y_i^2)}.$$

For fixed $Y = \prod_i y_i > 0$, let $z_i = \log y_i$. Since $\phi(z) = \log(1+e^{2z})$ is strictly convex, Jensen's inequality shows that the denominator is minimised at $z_1 = \cdots = z_n$, hence $y_1 = \cdots = y_n$. If $Y = 0$, setting every $y_i = 0$ minimises the denominator. This proves (5) globally, including boundaries.

A.4 Closed forms and first-order transition

For common y, the nonconstant part of the support is

$$g_n(y) = \frac{1-\lambda+\lambda 2^{n-1}y^n}{(1+y^2)^n}.$$

Direct maximisation gives the one-slot circle and (6). For $n \geq 3$,

$$g_n'(y) = \frac{ny}{(1+y^2)^{n+1}}\left[\lambda 2^{n-1}y^{n-2}(1-y^2) - 2(1-\lambda)\right].$$

The endpoint $y = 0$ remains locally maximal. Requiring a nonzero stationary point to tie it eliminates λ and yields (7); substituting back gives (8). On the interval in (7), the left-hand side decreases strictly from its maximum, which exceeds one, to zero; hence exactly one root exists. Since the tying y is nonzero, the exposed optimal strength jumps. Expanding around $z_n = 1$ gives

$$1 - z_n = 2^{1-n} + \frac{n-1}{2}2^{2-2n} + O(n^2 2^{3-3n}),$$
$$\lambda_c(n) = \frac{2}{3} - \frac{2^{1-n}}{9} + O(n 2^{2-2n}).$$

A.5 Collective classical-record optimum

Let $D = 2^n$ and $d = D/2$. For a collective outcome k, set

$$a_k = \sum_{x:\mathrm{par}(x)=0} p(k|x), \qquad b_k = \sum_{x:\mathrm{par}(x)=1} p(k|x).$$

Here $\mathrm{par}(x) = x_1 \oplus \cdots \oplus x_n$. Optimal parity decoding gives

$$P_\mathrm{A} = \frac{1}{2} + \frac{1}{2D}\sum_k |a_k - b_k|.$$

The recovery lemma followed by Cauchy–Schwarz within each parity sector gives

$$F_\mathrm{R} \leq \frac{1}{D^2}\sum_k (\sqrt{da_k} + \sqrt{db_k})^2.$$

With $A_k = a_k/d$, $B_k = b_k/d$,

$$T = \frac{1}{2}\|A - B\|_1, \qquad C = \sum_k \sqrt{A_k B_k},$$

these definitions give $P_\mathrm{A} = (1+T)/2$ and $F_\mathrm{R} \leq (1+C)/2$. The classical fidelity–total-variation relation $T^2 + C^2 \leq 1$ [15] yields $F_\mathrm{R} \leq f(T)$. Optimising the resulting one-variable score proves (9). A weak measurement of the two parity projectors attains equality.

Finally, for any $n \geq 2$, $\prod_i f(t_i) \leq f(\prod_i t_i)$. The collective device can therefore reproduce the same parity bias with no greater disturbance and is strictly better for every nontrivial interior support direction. This proves the strict resource gap.

B Reproducibility specification

The release artifact uses Python 3.10.5, NumPy 2.2.6, SciPy 1.15.3, Matplotlib 3.10.9, pytest 9.1.1, and Tectonic 0.17.0. From the repository root:

```
python -m pip install -e ".[dev,reproducibility]"
python -m pytest -q
python scripts/regenerate.py \
  --seed 20260812 --shots 8192
python scratch/originality_gate/\
  validate_streaming_parity.py
python scripts/build_pdf.py
```

The committed outputs include `data/audit_return_frontier.csv`, with 606 exact support rows spanning six stream lengths and 101 weights, and `data/audit_return_forecast.csv`, with 200 planning rows under the declared example law. The corresponding vector figures are `audit_return_frontier.pdf` and `audit_return_benchmark.pdf`. Stored adversarial searches appear in `streaming_parity.json`, while `streaming_parity_validation.json` records the fast independent validation summary.

Timestamps in generated figure PDFs are suppressed to make figure generation deterministic within a fixed environment. The manuscript PDF records its build time. Raster and vector rendering can differ across operating systems because of font and rendering-library versions; numerical CSV drift is the stronger cross-platform invariant.

The forecast does not simulate a specific device's density matrix. It evaluates the exact theorem and statistical planner under (12). Experimental use must replace its probabilities with measured or independently calibrated values.

C Claim ledger

The exact allowed novelty sentence is:

> To the author's knowledge, for the precisely specified streaming class, this work derives the exact parity-audit versus global-entanglement-return support, including arbitrary adaptive non-QND instruments, and proves a strict same-task separation from collective classical-record strategies and from a one-qubit coherent accumulator.

The sentence must be revised if priority review identifies a direct antecedent.

Table 2: Permitted claims and their evidence status at the 12 August 2026 cutoff.

Claim	Status	Boundary
Equation (5) is the exact adaptive classical-memory support	Proved here	Requires the factorised classical-path null and trusted access model
The balanced null is 3/4 for $n \geq 2$	Corollary	Uses global RETURN fidelity
The $n \geq 3$ exposed optimum has a first-order transition	Proved here	A decision transition of the optimiser, not a material phase transition
Collective classical recording obeys (9)	Proved here	The joint instrument may be coherent, but the device retains only a classical outcome after commitment
One persistent coherent qubit attains unit score	Exact circuit	Assumes ideal gates, source, storage, and decoder
The fixed-sample rule has false-positive probability at most α	Proved from Hoeffding	Independent Bernoulli trials, preregistered sample sizes, and a preregistered branch weight
Under the default forecast, certification remains feasible through $n = 143$	Numerical forecast	Illustrative model, not hardware calibration
No exact prior theorem was located	Dated search result	Not proof of priority; subject to specialist review
A positive experiment certifies a non-classical cross-slot resource	Model-relative	Rejects the declared classical-memory streaming null; attribution requires an interface audit
The protocol deletes every record or supports an interpretation	Not claimed	Neither follows from the score

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